Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

In a classical model of the tritium atom with a nucleus of charge $+I$ and a single electron in a circular orbit of radius r_0, suddenly the nucleus emits an electron and changes to charge $+2$ (The emitted electron escapes rapidly and we can forget about it). The electron in orbit suddenly has a neiv situation. Find the distance of closest approach for the new orbit in terms of r_0.

(A) $\frac{r_0}{2}$

(B) $\frac{r_0}{3}$

(C) r_0

(D) None of these

Q.2

The electric potential between a proton and an electron is given by $V = V_0 \ln \frac{r}{r_0}$, where r_0 is a constant. Assuming Bohr's model to be applicable, then variation of r_n with n, where n being the principal quantum number is

(A) $r_n \propto n$

(B) $r_n \propto \frac{1}{n}$

(C) $r_n \propto n^2$

(D) $r_n \propto \frac{1}{n^2}$

Q.3

In a certain hypothetical radioactive decay process, species A decays into species B and species B decays into species C according to the reactions

$A \longrightarrow 2B + \text{ particles} + \text{energy:} \quad B \longrightarrow 3C + \text{ particles} + \text{energy}$

The decay constant for the species A is $\lambda_1 = 1\,\mathrm{sec}^{-1}$ and that for the species B is $\lambda_2 = 100\mathrm{sec}^{-1}$. Initially 10^4 moles of the species of A were present while there was none of B and C. It was found that species B reaches its maximum number at a time $t_0 = 2\ln(10)$ sec. Then value of the maximum number of moles of B is $\left(e^{2.303} = 10\right)$

(A) 2

(B) 3

(C) 4

(D) 6

Q.4

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A radioactive nucleus decays according to following series

$$X \xrightarrow{\alpha} X_1 \xrightarrow{\beta} X_2 \xrightarrow{\alpha} X_3 \xrightarrow{\gamma} X_4$$

If the atomic number and atomic weight of the parent element X are 72 and 180 respectively, then the atomic number and atomic mass of X_4 are respectively

(A) 70,172

(B) 69,171

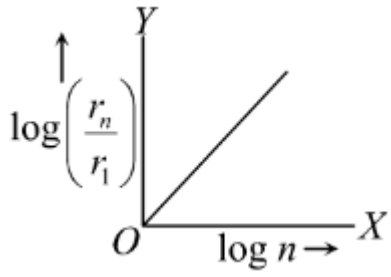
(C) 69,172

(D) 68,172

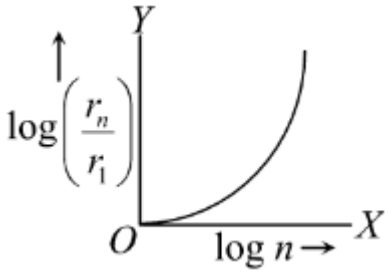
Q.5

In hydrogen atom, the radius of nth Bohr's orbit is r_n. The graph between $\log\left(\frac{r_n}{r_1}\right)$ and $\log n$ will be

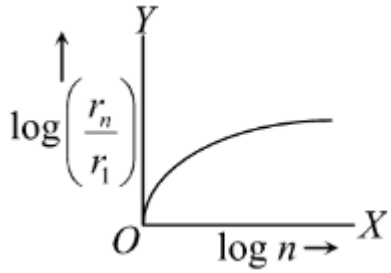
(A)



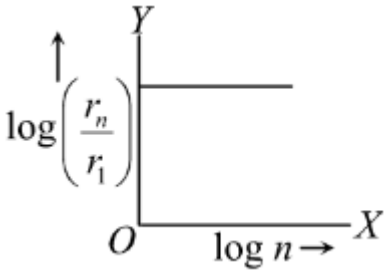
(B)



(C)



(D)



Q.6

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When a monochromatic $3\ \mathrm{W}$ source is at $0.3\ \mathrm{m}$ from a photoelectric cell, the cut-off voltage and saturation current are respectively $0.3\ \mathrm{V}$ and $3\ \mathrm{mA}$. If the power of the source is reduced to $1\ \mathrm{W}$ and its distance from the cell is reduced to $0.1\ \mathrm{m}$ then -

(A) stopping potential will be $0.1\ \mathrm{V}$

(B) saturation potential will be $3\ \mathrm{mA}$

(C) stopping potential will be $0.9\ \mathrm{V}$

(D) saturation current will be $27\ \mathrm{mA}$

Q.7

If light of wavelength of maximum intensity emitted from surface at temperature T_1 is used to cause photoelectric emission from a metallic surface, the maximum kinetic energy of the emitted electron is $6\mathrm{ev}$, which is 3 times the work function of the metallic surface. If light of wavelength of maximum intensity emitted from a surface at temperature $T_2\ (T_2 = 2T_1)$ is used, the maximum kinetic energy of the: photoelectrons emitted is

(A) $14\mathrm{eV}$

(B) $4\mathrm{eV}$

(C) $2\mathrm{eV}$

(D) $18\mathrm{eV}$

Q.8

A hydrogen like atom in an excited state n emits a wavelength of Balmer series. n cannot be -

(A) 6

(B) 7

(C) 8

(D) 9

Q.9

Light of wavelength of $2000\mathring{A}$ falls on an aluminium surface. For aluminium, $4.2\mathrm{eV}$ is required to remove an electron from its surface. Then, (Planck's constant $\mathrm{h} = 6.6 \times 10^{-34}\ \mathrm{J.s}$ and speed of light $= 3 \times 10^{8}\ \mathrm{m/s}$)

(A) The kinetic energy of the fastest photoelectrons is $2\mathrm{eV}$

(B) by doubling the wavelength of light the maximum kinetic energy of photoelectrons is $1\mathrm{eV}$.

(C) stopping potential is $2\ \mathrm{V}$

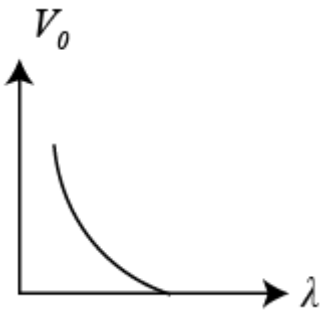
(D) cut off wavelength for aluminium is $3000\mathring{A}$

Q.10

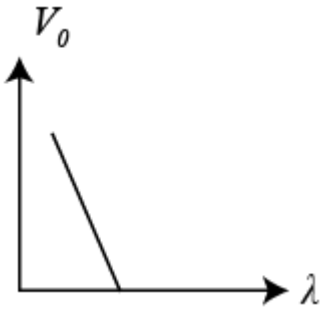
For photo-electric effect with incident photon wavelength λ the stopping potential is V_0 identify the correct variation(s) of V_0 with λ or $\frac{1}{\lambda}$

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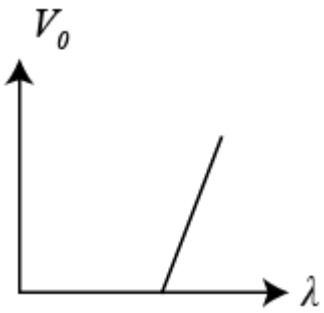
(A)



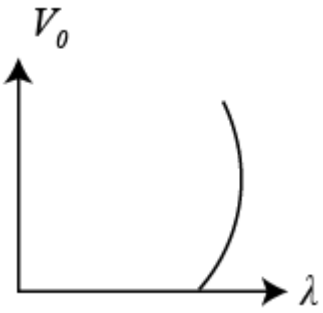
(B)



(C)



(D)



Q.11

Binding energy per electron are $\mathrm{E}_1, \mathrm{E}_2$ and E_3 respectively for hydrogen atom, He atom and Li^+ ion, all being at ground state. Then -

(A) $\mathrm{E}_1 > \frac{\mathrm{E}_2}{4}$

(B) $\mathrm{E}_1 > \frac{\mathrm{E}_3}{9}$

(C) Energy required to singly ionise a He atom in ground state is $4\mathrm{E}_1$

(D) Energy required to singly ionise a He atom in ground state is E_2

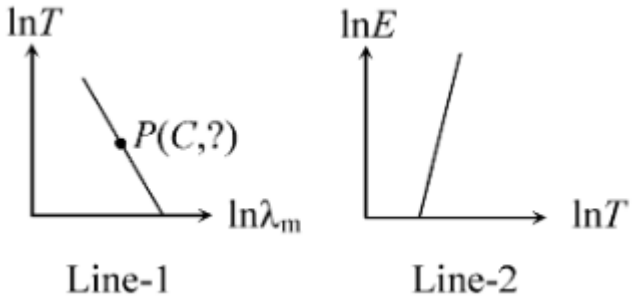
Q.12

Paragraph

The variation of the $\ln T$ versus $\ln \lambda_{\mathrm{m}}$ and $\ln E$ versus $\ln T$ are given as shown. T is the temperature of the body in Kelvins, λ_m is the wave length corresponding to which maximum number of photons are emitted and E is the

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energy emitted by the body per second. The intercept made by the line 1 on the Y-axis has length A.



Question

What is the slope of the line 1 ?

(A) $-1/2$

(B) -1

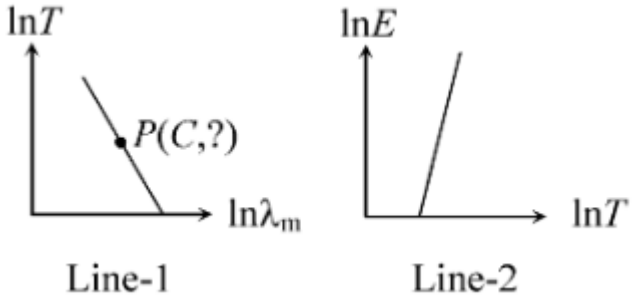
(C) -2

(D) -4

Q.13

Paragraph

The variation of the $\ln T$ versus $\ln \lambda_m$ and $\ln E$ versus $\ln T$ are given as shown. T is the temperature of the body in Kelvins, λ_m is the wave length corresponding to which maximum number of photons are emitted and E is the energy emitted by the body per second. The intercept made by the line 1 on the Y-axis has length A.



Question

What is the slope of the line $2?$

(A) $1/2$

(B) 1

(C) 2

(D) 4

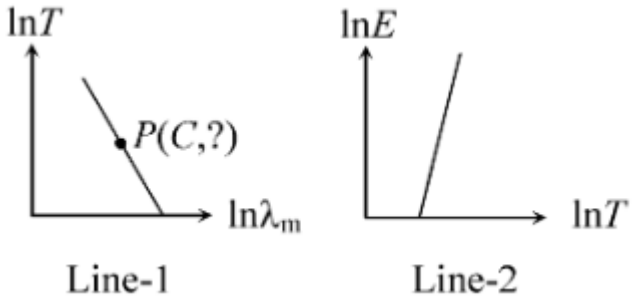
Q.14

Paragraph

The variation of the $\ln T$ versus $\ln \lambda_m$ and $\ln E$ versus $\ln T$ are given as shown. T is the temperature of the body in Kelvins, λ_m is the wave length corresponding to which maximum number of photons are emitted and E is the

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energy emitted by the body per second. The intercept made by the line 1 on the Y-axis has length A.



Question

What is the value of Wein's displacement constant?

(A) e^{A}

(B) $1/e^{A}$

(C) $\ln A$

(D) $1/\ln A$

Q.15

Passage

Work function of metal A is equal to the ionization energy of hydrogen atom in first excited state. Work function of metal B is equal to the ionization energy of He^{+}ion in second orbit. Photons of same energy E are incident on both A and B. Maximum kinetic energy of photoelectrons emitted from A is twice that of photoelectrons emitted from B .

Question

Value of E (in eV) is :

(A) 20.8

(B) 32.2

(C) 24.6

(D) 23.8

Q.16

Passage

Work function of metal A is equal to the ionization energy of hydrogen atom in first excited state. Work function of metal B is equal to the ionization energy of He^{+}ion in second orbit. Photons of same energy E are incident on both A and B. Maximum kinetic energy of photoelectrons emitted from A is twice that of photoelectrons emitted from B .

Question

The difference in maximum kinetic energy of photoelectrons from A and from B :

(A) increases with increase in E

(B) decreases with increase in E

(C) first increases than decrease with increase in E

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(D) remain constant

Q.17

Passage

Work function of metal A is equal to the ionization energy of hydrogen atom in first excited state. Work function of metal B is equal to the ionization energy of He^{+}ion in second orbit. Photons of same energy E are incident on both A and B. Maximum kinetic energy of photoelectrons emitted from A is twice that of photoelectrons emitted from B .

Question

For a photon of energy h v to eject an electron from a material with a work function ϕ, the photon's energy must be

(A) greater than ϕ

(B) less than ϕ

(C) less than maximum kinetic energy

(D) equal to maximum kinetic energy

Q.18

Natural samarium (atomic weight $=\mathrm{W}=150.43$ grams) has been found to emit α particle at a rate of 135 particles per gram per second.

The isotope $\mathrm{S_m}$ of mass number $\mathrm{A}=147$ and atomic number $\mathrm{z}=62$, and abundance 15% is responsible for the activity

Column I	**Column II**
(A) probability of a given nucleus of $_{62}\mathrm{Sm}^{147}$ disintegrating in the next one second	(p) independent of W
(B) half-life of $_{62}\mathrm{Sm}^{147}$	(q) independent of mass number A
(C) specific activity (in disintegration per gram per second) after one year	(r) independent of z
(D) Q value of the decay process	(s) independent of abundance (15%)

(A) $(\mathrm{A})-\mathrm{p},(\mathrm{B})-\mathrm{r}, \mathrm{q},(\mathrm{C})-\mathrm{s}, \mathrm{r},(\mathrm{D})-\mathrm{q}$

(B) $(\mathrm{A})-\mathrm{s}, \mathrm{p}(\mathrm{B})-\mathrm{p},(\mathrm{C})-\mathrm{s},(\mathrm{D})-\mathrm{q}$

(C) $(\mathrm{A})-\mathrm{q}, \mathrm{r}(\mathrm{B})-\mathrm{q}, \mathrm{r}(\mathrm{C})-\mathrm{p}, \mathrm{q}, \mathrm{r}, \mathrm{s}(\mathrm{D})-\mathrm{p}, \mathrm{s}$

(D) $(\mathrm{A})-\mathrm{p}, \mathrm{q}, \mathrm{r}(\mathrm{B})-\mathrm{q},(\mathrm{C})-\mathrm{r},(\mathrm{D})-\mathrm{p}$

Q.19

A target element A is bombarded with electrons and the wavelengths of the characteristic spectrum are measured. A second characteristic spectrum is also obtained, because of an impurity in the target. The wavelength of the K_a lines are $196\mathrm{pm}$ (element A) and $169\mathrm{pm}$ (impurity). If the atomic number of impurity is z $=(10x-1)$. Find the value of x. (atomic number of element A is 27).

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Q.20

About $185\mathrm{MeV}$ of usable energy is released in the neutron-induced fissioning of a $^{235}_{92}\mathrm{U}$ nucleus. $^{235}_{92}\mathrm{U}$ in a reactor is continuously generating $100\mathrm{MW}$ of power. If the number of days that it will take for $1\ \mathrm{kg}$ of the uranium to be used up is $\mathrm{n} \times 10^5$, find nearest integral value of n.

Answers & Solutions

Q.1 Answer:

$\frac{r_0}{3}$

Solution:

Before emission :

$$\frac{mv_0^2}{r_0} = \frac{e^2}{4\pi\varepsilon_0 r_0^2}$$

$$\mathrm{KE} = \frac{\mathrm{m}v_0^2}{2} = \frac{\mathrm{e}^2}{8\pi\varepsilon_0 \mathrm{r}_0}$$

After emission : $KE =$ **same**

$$\mathrm{PE} = -\frac{2\mathrm{e}^2}{4\pi\varepsilon_0 \mathrm{r}_0}$$

$$\mathrm{E}_2 = \frac{mv_0^2}{2} - \frac{2\mathrm{e}^2}{4\pi\varepsilon_0 \mathrm{r}} = -\frac{3\mathrm{e}^2}{8\pi\varepsilon_0 \mathrm{r}_0}$$

$$\frac{E_2}{E_1} = 3$$

Finally velocity is perpendicularly $\mathrm{mv}_0\mathrm{r}_0 = \mathrm{mvr}$

and $\frac{-3\mathrm{e}^2}{8\pi\varepsilon_0 \mathrm{r}_0} = \frac{1}{2}\mathrm{mv}^2 - \frac{\mathrm{e}^2}{2\pi\varepsilon_0 \mathrm{r}}$

After solving $\mathrm{r} = \frac{\mathrm{r}_0}{3}$

Q.2 Answer:

$r_n \propto n$

Solution:

$\mathrm{E} = \frac{F}{e} = \frac{-dV}{dr} = -\frac{V_0}{r}, \quad |F| = \frac{V_0 e}{r} = \frac{mv^2}{r} \Rightarrow v \propto r^0, \quad v = \sqrt{\frac{v_0 e}{m}}, \quad mvr_n = \frac{nh}{2\pi}$

Velocity is constant hence $r_n \propto n$

$\therefore$ **(a)**

Q.3 Answer:

2

Solution:

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$$\frac{dN_A}{dt} = -\lambda_1 N_A, \frac{dN_B}{dt} = 2\lambda_1 N_A - \lambda_2 N_B,$$

$N_B =$ maximum $\Rightarrow \frac{dN_B}{dt} = 0$

$\Rightarrow \quad 2\lambda_1 N_A = \lambda_2 N_{B_{\max}}$

$\Rightarrow \quad N_{B_{\max}} = \frac{2\lambda_1}{\lambda_2} N_A$

$\Rightarrow \quad N_{B_{\max}} = \frac{2\lambda_1}{\lambda_2} N_0 e^{-\lambda_1 t} = 2.$

$\therefore \quad$ (A)

Q.4 Answer:

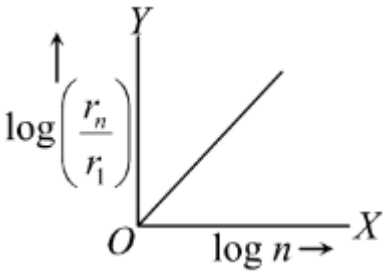
69,172

Solution:

$^{180}_{72}X \xrightarrow{\alpha} {}^{176}_{70}X_1 \xrightarrow{\beta} {}^{176}_{71}X_2 \xrightarrow{\alpha} {}^{172}_{69}X_3 \xrightarrow{\gamma} {}^{172}_{69}X_4$

$\therefore$ **(c)**

Q.5 Answer:



Solution:

$r_n \propto n^2; \quad \log \frac{r_n}{r_1} = 2 \log n$

$\therefore \quad$ (a)

Q.6 Answer:

saturation current will be $27\ \mathrm{mA}$

Solution:

Wavelength same $\Rightarrow h\nu - \phi$ remains same $\Rightarrow$ no change in stopping potential. Power of source becomes $\frac{1}{3}$, results in : intensity becomes $\frac{1}{3}$, distance becomes $\frac{1}{3}$, result in : intensity becomes 9 times

$\Rightarrow$ net effect intensity becomes 3 times

$\Rightarrow$ saturation current becomes 9 times

Q.7 Answer:

14eV

Solution:

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$$\text{Case 1}: \frac{hc}{\lambda_1} = 2\text{eV} + 6\text{eV}$$

$$\Rightarrow \lambda_1 = \frac{hc}{8\text{eV}}$$

$$\text{Case 2}: \frac{hc}{\lambda_2} = 2\text{eV} + KE_{\max}$$

$$\Rightarrow \frac{\lambda_2}{\lambda_1} = \frac{T_1}{T_2} = \frac{T}{2T} = \frac{1}{2} \left[\text{Wien's Displacement Law}, \lambda_m \propto \frac{1}{T} \right]$$

$$\lambda_2 = \frac{\lambda_1}{2}$$

$$\Rightarrow \lambda_2 = \frac{hc}{16\text{eV}}$$

$$\therefore 16\text{eV} = 2\text{eV} + KE_{\max}$$

$$\Rightarrow KE_{\max} = 14\text{eV}$$

Q.8 Answer:

7

Solution:

For hydrogen, $\frac{1}{\lambda} = \mathrm{R}\left(\frac{1}{\mathrm{n}_1^2} - \frac{1}{\mathrm{n}_2^2}\right)$

For Balmer series, $\mathrm{n}_1 = 2, \mathrm{n}_2 =$ **any** > 2

For hydrogen like, $\frac{1}{\lambda} = \mathrm{R}\left[\frac{1}{\left(\frac{n_1}{z}\right)^2} - \frac{1}{\left(\frac{n_2}{z}\right)^2}\right]$

Given, $\frac{\mathrm{n}_1}{\mathrm{z}} = 2$ **and** $\frac{\mathrm{n}_2}{\mathrm{z}} =$ **any,**> 2

possibilities : For $\mathrm{z} = 2, \mathrm{n}_1 = 4$ **and** $\mathrm{n}_2 = 6, 8, 10, \ldots\ldots\ldots\ldots$

for $z = 3, n_1 = 6$ **and** $n_2 = 9, 12, \ldots\ldots\ldots$

$\therefore$ n_2 **cannot be 7**

Q.9 Answer:

The kinetic energy of the fastest photoelectrons is 2eV

Solution:

Energy corresponding to incident photon $= 9.9 \times 10^{-19}$ **Joules** $= \frac{9.9 \times 10^{-19}}{1.6 \times 10^{-19}}\text{eV} = 6.2\text{eV}$

The kinetic energy of fastest photoelectrons $\mathrm{E_k} = \mathrm{h}v - \mathrm{W} = 6.2\text{eV} - 4.2\text{eV} = 2\text{eV}$ **If** $\mathrm{V_s}$ **is the stopping potential, then** $\mathrm{E_k} = \mathrm{eV_s}$

$\mathrm{V_s} = \frac{\mathrm{E_k}}{\mathrm{e}} = \frac{2\text{eV}}{\mathrm{e}} = 2\ \mathrm{V}$

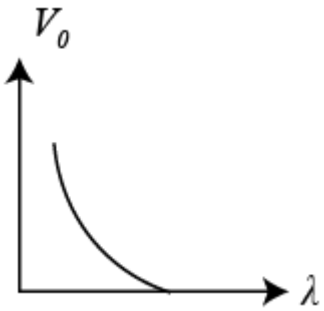
If λ_0 **is cut off wavelength for aluminium, then** $W = \left(\frac{hc}{\lambda_0}\right)$

$\lambda_0 = \frac{\mathrm{hc}}{\mathrm{W}} = \frac{6.8 \times 10^{-34} \times 3 \times 10^8}{4.2 \times 1.6 \times 10^{-19}} = 3000 \times 10^{-10}\ \mathrm{m} = 3000\mathring{A}$

$\therefore$ By doubling the wavelength, no photoelectrons are emitted.

Q.10 Answer:

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Solution:

$$V_0 e = \frac{hC}{\lambda} - w_0$$

$$V_0 = \left(\frac{hC}{e}\right)\left(\frac{1}{\lambda}\right) - \frac{w_0}{e}$$

$V_0, \frac{1}{\lambda}$ graph is in the from of $y = mx - c$

Option C is correct.

Clearly, V_0, λ graph is a curve. V_0 decreases with increase in λ V_0 becomes zero when $\frac{hC}{\lambda} = w_0$ i.e. $\lambda = \lambda_0$ (Threshold wavelength)

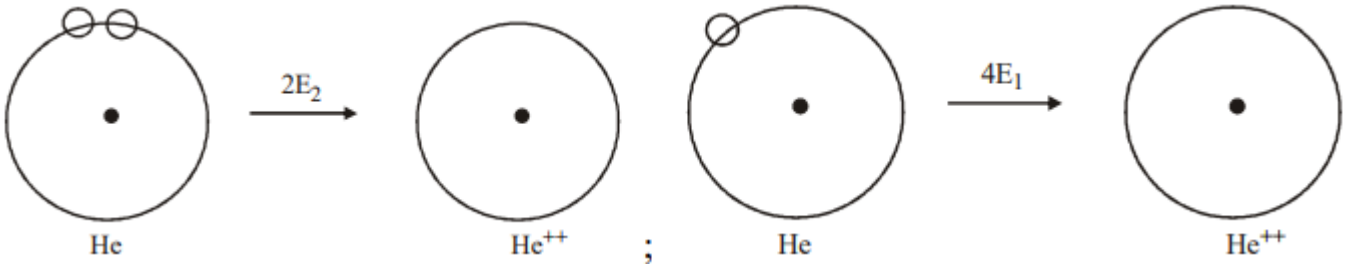
Option A is correct.

Q.11 Answer:

$\mathrm{E}_1 > \frac{\mathrm{E}_2}{4}$

Solution:

Given :



We know, binding energy of the second electron in He say x will be less than $4E_1$ (i.e. (C) is incorrect)

$\therefore \quad 2\mathrm{E}_2 < 8\mathrm{E}_1 \Rightarrow \mathrm{E}_1 > \frac{\mathrm{E}_2}{4}$. **Similarly,** $\mathrm{E}_1 > \frac{\mathrm{E}_3}{9}$

Also, $x + 4E_1 = 2E_2 \Rightarrow x = 2E_2 - 4E_1 = E_2 + E_2 - 4E_1$

But $\mathrm{E}_2 - 4\mathrm{E}_1 < 0 \quad \therefore \mathrm{x} > \mathrm{E}_2$

Q.12 Answer:

-1

Solution:

$T\lambda_m = b \Rightarrow \ln T = \ln b - \ln \lambda_m$

$\therefore \quad \text{slope} = -1$

$\therefore \quad (\mathrm{b})$

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Q.13 Answer:

4

Solution:

$$E = e\sigma AT^4 \Rightarrow \ln E = \ln(\rho\sigma A) + 4\ln T$$
$$\therefore \quad \text{slope } = 4$$
$$\therefore \quad (\text{d})$$

Q.14 Answer:

e^A

Solution:

$$\ln T = \ln b - \ln \lambda_m$$
$$\text{when } \ln \lambda_m = 0, \ln T = A$$
$$\Rightarrow \quad \ln b = A \Rightarrow b = e^A$$
$$\therefore \quad (\text{a})$$

Q.15 Answer:

23.8

Solution:

$$\phi_{\mathrm{A}} = \frac{13.6}{4}\mathrm{eV}, \phi_{\mathrm{B}} = \frac{13.6\times 4}{4}\mathrm{eV}$$
$$\mathrm{E} - \phi_{\mathrm{A}} = (\mathrm{KE})_{\mathrm{A}}; \mathrm{E} - \phi_{\mathrm{B}} = (\mathrm{KE})_{\mathrm{B}}$$
$$\mathrm{KE}_{\mathrm{A}} = 2\mathrm{KE}_{\mathrm{B}}$$
$$\mathrm{E} - \phi_{\mathrm{A}} = 2\left(\mathrm{E} - \phi_{\mathrm{B}}\right)$$
$$\mathrm{E} = 2\phi_{\mathrm{B}} - \phi_{\mathrm{A}} = 2 \times 13.6 - \frac{13.6}{4} = 23.8\mathrm{eV}$$

Q.16 Answer:

remain constant

Solution:

$$\phi_{\mathrm{A}} = \frac{13.6}{4}\mathrm{eV}, \phi_{\mathrm{B}} = \frac{13.6\times 4}{4}\mathrm{eV}$$
$$\mathrm{E} - \phi_{\mathrm{A}} = (\mathrm{KE})_{\mathrm{A}}; \mathrm{E} - \phi_{\mathrm{B}} = (\mathrm{KE})_{\mathrm{B}}$$
$$\mathrm{KE}_{\mathrm{A}} = 2\mathrm{KE}_{\mathrm{B}}$$
$$\mathrm{E} - \phi_{\mathrm{A}} = 2\left(\mathrm{E} - \phi_{\mathrm{B}}\right)$$
$$\mathrm{E} = 2\phi_{\mathrm{B}} - \phi_{\mathrm{A}} = 2 \times 13.6 - \frac{13.6}{4} = 23.8\mathrm{eV}$$

Q.17 Answer:

greater than ϕ

Solution:

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$\phi_{\mathrm{A}}=\frac{13.6}{4} \mathrm{eV}, \phi_{\mathrm{B}}=\frac{13.6 \times 4}{4} \mathrm{eV}$
$\mathrm{E}-\phi_{\mathrm{A}}=(\mathrm{KE})_{\mathrm{A}} ; \mathrm{E}-\phi_{\mathrm{B}}=(\mathrm{KE})_{\mathrm{B}}$
$\mathrm{KE}_{\mathrm{A}}=2 \mathrm{KE}_{\mathrm{B}}$
$\mathrm{E}-\phi_{\mathrm{A}}=2\left(\mathrm{E}-\phi_{\mathrm{B}}\right)$
$\mathrm{E}=2 \phi_{\mathrm{B}}-\phi_{\mathrm{A}}=2 \times 13.6-\frac{13.6}{4}=23.8 \mathrm{eV}$

Q.18 Answer:

$(\mathrm{A})-\mathrm{q}, \mathrm{r}(\mathrm{B})-\mathrm{q}, \mathrm{r}(\mathrm{C})-\mathrm{p}, \mathrm{q}, \mathrm{r}, \mathrm{s}(\mathrm{D})-\mathrm{p}, \mathrm{s}$

Solution:

$(A)-\mathbf{qr},(B)-\mathbf{qr},(C)-\mathbf{pqrs},(D)-\mathbf{ps}.$

Probability $=\lambda$

$-\frac{\mathrm{dN}}{\mathrm{dt}}=\mathrm{N} \lambda$

Number of nuclei of ${}_{62}\mathrm{Sm}^{147}$ **per gram of samarium**

$$= \mathrm{N}=\frac{\text{Avagadro no.}}{\mathrm{W}} \times \text{ abundance } =\frac{6.023 \times 10^{23}}{150.43} \times 0.15=6 \times 10^{20}$$

$$\text{Given } -\frac{\mathrm{dN}}{\mathrm{dt}}=135 \quad \therefore \lambda=\frac{135}{\mathrm{N}}, \text{ depends on N}$$

$\Rightarrow$ depends on W and abundance independent of A and Z. Since half life $=\dfrac{0.693}{\lambda}$

half life also is independent of A and Z.

$\text{Activity } =\mathrm{A}_0 \mathrm{e}^{-\lambda \mathrm{t}} \mathrm{A}_0(\because \lambda \text{ is very small, } \mathrm{t}=1 \text{ year } \Rightarrow \lambda \mathrm{t} \text{ is very small })$

${}_{62}\mathrm{Sm}^{147} \rightarrow {}_{60}\mathrm{X}^{143}+{}_{2}\mathrm{He}^{4}+\mathrm{Q}$

Q does not depend on W and abundance (but depends on mass of Sm^{147} **which in turn depends on mass of consisting (A and Z) and on mass defect)**

Q.19 Answer:

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Solution:

$Z-1=\sqrt{\frac{4 v}{3 c R}}$
$z-1 \propto \sqrt{\frac{1}{\lambda}}$
$\frac{z_2-1}{z_1-1}=\sqrt{\frac{\lambda_1}{\lambda_2}}=\sqrt{\frac{196}{169}}=\frac{14}{13}$
$z_2-1=26 \times \frac{14}{13}=2 B$
$\mathbf{z}_2=\mathbf{29}=10 \pi-1$
$\mathbf{x}=\mathbf{3}$

Q.20 Answer:

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Solution:

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The fission rate corresponding to the given power output is (i.e. no. of fissions/sec)

$\left(10^{8}\ \mathrm{J/s}\right)\left(\frac{10^{-6}\mathrm{MeV}}{1.6\times10^{-19}\ \mathrm{J}}\right)\left(\frac{1\text{ fission}}{185\mathrm{MeV}}\right)$

$3.38\times10^{18}\text{ fissions }/\mathrm{s}=3.4\times10^{18}\text{ fissions }/\mathrm{s}$

No. of nuclei in 1 kg of $^{235}\mathrm{U}=\left(\frac{1}{235}\right)\left(6.023\times10^{26}\right)=2.56\times10^{24}$

$\Rightarrow$ **Required minimum numbers of days** $=\frac{2.56\times10^{24}}{3.4\times10^{18}}=0.752\times10^{6}=7.52\times10^{5}=8\times10^{5}$

$\Rightarrow \mathrm{n}=8$

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